

Black Hole Information Paradox:

0/0 at the Event Horizon

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Abstract

We show that the black hole information paradox has a 0/0 removable singularity structure at the event horizon. At the horizon, information inside = 0 and information outside = infinity, giving 0/0. The removable value is the Bekenstein-Hawking entropy $S_{BH} = A/(4l_{Planck}^2)$. This connects to the Toomre Q parameter ($Q=1$ at horizon, $\beta=1/2$), the holographic principle (information $\sim$ area), and the Page curve (information recovery at t_{Page}).

1. Introduction

The black hole information paradox (Hawking 1975) asks: what happens to information that falls into a black hole? Hawking radiation appears to destroy information, violating quantum unitarity. We show that the paradox has a 0/0 removable singularity structure at the event horizon.

2. The 0/0 at the Event Horizon

At the event horizon ($r = r_s$):

- Information inside: $I = 0$ (nothing escapes)
- Information outside: $I = \text{infinity}$ (all information)
- 0/0: $I_{inside} / I_{outside} = 0 / \text{infinity} = 0$

Removable value: $S_{BH} = A / (4 * l_{Planck}^2)$

This is the Bekenstein-Hawking entropy. The event horizon is a 0/0 removable singularity in the space of quantum information.

3. Bekenstein-Hawking Entropy

$$S_{BH} = k_B * A / (4 * l_{Planck}^2)$$

For a solar mass black hole: $S_{BH}/k_B \sim 10^{77}$. For Sgr A*: $S_{BH}/k_B \sim 10^{90}$. For M87*: $S_{BH}/k_B \sim 10^{104}$.

4. Page Curve

The Page curve describes information recovery during evaporation:

- $t = 0$: $I = 0$ (no radiation)
- $t = t_{Page}$: $I = S_{BH}$ (Page time)
- $t = t_{evap}$: $I = S_{BH}$ (all information recovered)

The 0/0: $I(t_{Page}) = S_{BH}/1 = S_{BH}$ (removable). The Page time is when the radiation entropy equals $S_{BH}/2$.

5. Holographic Principle

$$\text{Maximum information} = \text{Area} / (4 * l_{Planck}^2)$$

The holographic principle states that the maximum information in a region is proportional to its AREA, not its VOLUME. This is MATHEMATICALLY EQUIVALENT to the Bekenstein-Hawking entropy.

Connection to Toomre Q: stability depends on SURFACE density, not volume density. Same 0/0 structure!

6. Connection to Toomre Q

The event horizon is a gravitational instability:

- $Q < 1$: unstable (horizon forms)
- $Q > 1$: stable (no horizon)
- $Q = 1$: marginal (0/0 removable singularity)

$$Q \sim (r - r_s)^{(1/2)}, \beta = 1/2 \text{ (mean-field Ising)}$$

The critical exponent $\beta = 1/2$ is the SAME as Toomre Q at $Q = 1$.

7. Main Results

THEOREM

Horizon 0/0: $I_{inside}/I_{outside} = 0/\text{infinity} = 0$, removable value $S_{BH} = A/(4l_{Planck}^2)$.

THEOREM

Page Curve: $I(t_{Page}) = S_{BH}$ (information recovery), 0/0 at t_{Page} .

THEOREM

Holographic: $S = A/(4l_{\text{Planck}}^2) = 0/0$ at horizon, removable S_{BH} .

8. Conclusion

The black hole information paradox has a $0/0$ removable singularity structure at the event horizon. The removable value is the Bekenstein-Hawking entropy. This connects to the Toomre Q parameter ($Q=1$ at horizon), the holographic principle (information $\sim$ area), and the Page curve (information recovery).

References

- [1] Bekenstein (1973). Phys. Rev. D 7, 2333.
- [2] Hawking (1975). Comm. Math. Phys. 43, 199.
- [3] Page (1993). Phys. Rev. Lett. 71, 1291.
- [4] Maldacena (1998). Adv. Theor. Math. Phys. 2, 231.
- [5] Penington (2020). JHEP 2020, 20.
- [6] Almheiri et al. (2021). Rev. Mod. Phys. 93, 35002.
- [7] Toomre (1964). ApJ 139, 1217.
- [8] Puno (2026). The Removable Singularity.